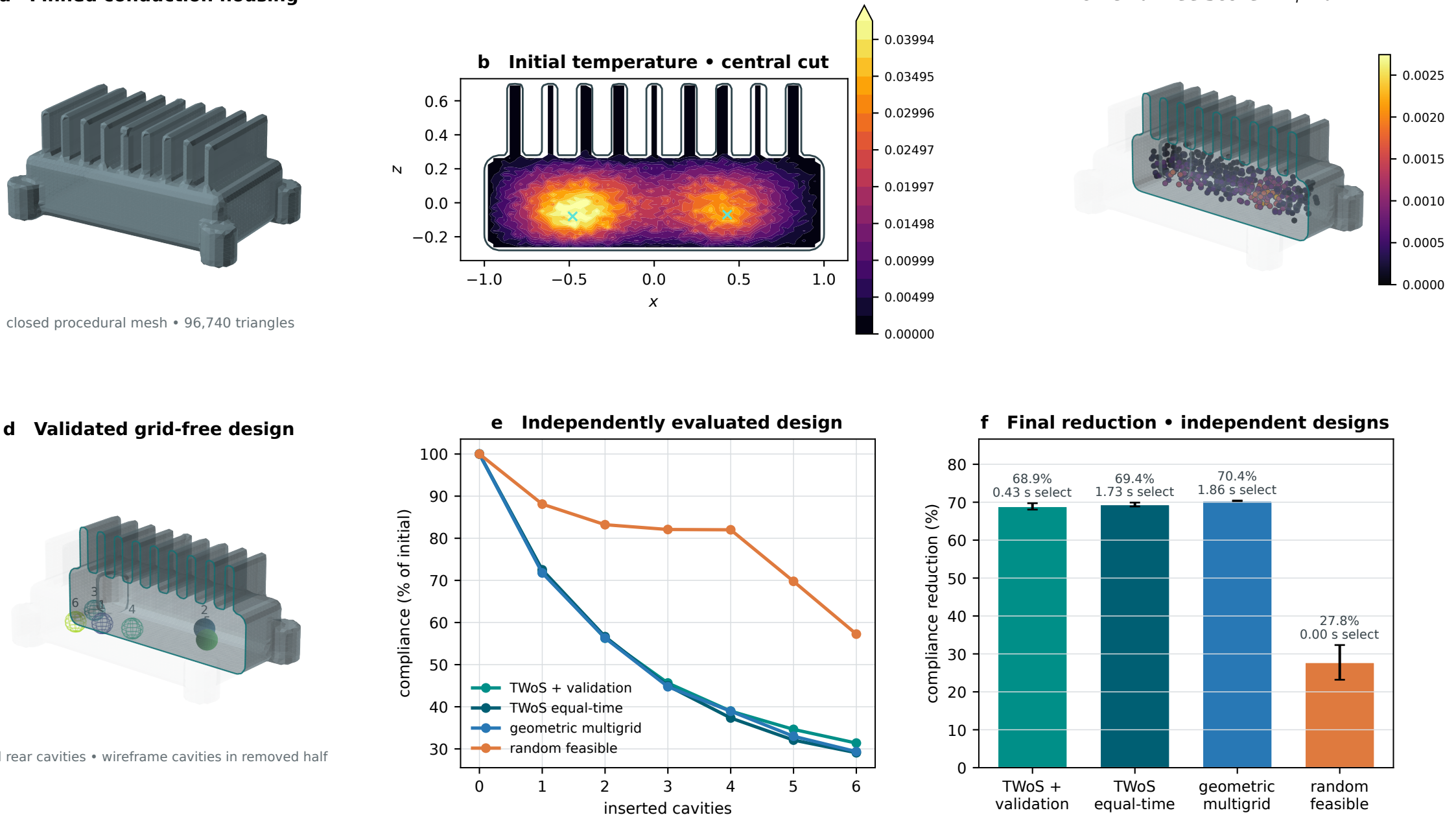

a Finned conduction housing
closed procedural mesh • 96,740 triangles
b Initial temperature • central cut
z
x
0.6
0.4
0.2
0.0
−0.2
−1.0
−0.5
0.0
0.5
1.0
0.03994
0.03495
0.02996
0.02497
0.01997
0.01498
0.00999
0.00499
0.00000
c Grid-free score $-D_T = u^2$
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
d Validated grid-free design
6
3
1
4
2
solid rear cavities • wireframe cavities in removed half
e Independently evaluated design
compliance (% of initial)
inserted cavities
100
90
80
70
60
50
40
30
0
1
2
3
4
5
6
TWoS + validation
TWoS equal-time
geometric multigrid
random feasible
f Final reduction • independent designs
compliance reduction (%)
80
70
60
50
40
30
20
10
0
68.9%
0.43 s select
69.4%
1.73 s select
70.4%
1.86 s select
27.8%
0.00 s select
TWoS + validation
TWoS equal-time
geometric multigrid
random feasible
stylized engineering geometry • localized volumetric heating • six radius-0.095 cavities • 8 CPU threads